Cryptocurrency Prediction and Market Analysis System

Software Requirement Specification done by

SHAIRAAJ V S - 25MX221

SURYA L - 25MX229

23MX41 – PROJECT WORK

Report submitted in partial fulfillment of the requirements for the degree of

MASTER OF COMPUTER APPLICATIONS

ANNA UNIVERSITY



AUGUST 2026

PSG COLLEGE OF TECHNOLOGY

(Autonomous Institution)

COIMBATORE – 641 004

Faculty Guide

…….

Dr. Subathra M

Assistant Professor (Sl. Grade)

Department of Computer Applications

PSG College of Technology

2) Student Detail

Roll No.	Name of the Student	Name of the Faculty Guide	Name of the Industry Guide, if any
25MX221	Shairaaj V S	Dr. M. Subathra, Assistant Professor (Sl. Gr.)	N.A.
25MX229	Surya L		

3) Project Goal

The goal of Cryptocurrency Prediction and Market Analysis System is to design and develop a web-based system that provides accurate cryptocurrency price predictions along with real-time market analysis. The system aims to assist users in making informed investment decisions by integrating machine learning techniques, sentiment analysis, and explainable AI. It focuses on improving transparency, usability, and predictive capability compared to existing platforms.

4) Project Abstract

The Cryptocurrency Prediction and Market Analysis System is designed to address the challenges faced by investors in understanding the highly volatile cryptocurrency market. Existing platforms mainly provide historical data and current prices but lack predictive analytics and insights into market sentiment.

This project introduces a web-based application that integrates machine learning and natural language processing to enhance decision-making. The system uses an XGBoost regression model to predict next-day cryptocurrency prices based on technical indicators such as moving averages, RSI, MACD, and volatility. In addition, it performs sentiment analysis on financial news using the FinBERT model to classify news as positive, negative, or neutral.

To improve user trust, the system incorporates SHAP (SHapley Additive Explanations), which explains how each feature contributes to the predicted price. The application is developed using React for the frontend, Flask for the backend, and SQLite for data storage.

The system provides a unified dashboard where users can view real-time prices, predictions, sentiment analysis, and explanatory insights. It is designed to be user-friendly and does not involve any blockchain transactions or trading features, making it safe and accessible for all users including beginners and researchers.

5) Existing System / Literature Review

Several existing platforms provide cryptocurrency-related information, but they have limitations. Platforms like **CoinGecko** and **CoinMarketCap** provide real-time price tracking and historical data visualization. However, they lack predictive analytics and sentiment-based insights.

Trading platforms such as **Binance** and **TradingView** offer technical analysis tools, but they require manual interpretation and do not provide automated predictions.

Decentralized prediction platforms like **Polymarket** and **Augur** allow users to bet on outcomes, but they focus on prediction markets rather than data-driven forecasting.

From literature, machine learning models such as LSTM and ARIMA have been used for price

prediction, but many lack integration with sentiment analysis.

Similarly, NLP-based sentiment analysis using Twitter or news data helps understand market trends but is rarely combined with price prediction models. Therefore, there is a need for a system that integrates: Price prediction, Sentiment analysis, Explainable AI.

S.No.	Existing System / Reference	Limitation Addressed
1	CoinGecko / CoinMarketCap	Only provides price tracking, no prediction or sentiment analysis
2	Trading/Charting Platforms	Requires manual analysis, no automated prediction
3	Polymarket / Augur	Focus on betting, not forecasting
4	NLP Sentiment Analysis models	Only analyzes sentiment, not linked with price prediction
5	Existing Crypto Tools	Lack of integrated system combining prediction + sentiment + visualization

Table 5.1 — Literature Review Summary

6) Project Flow Analysis

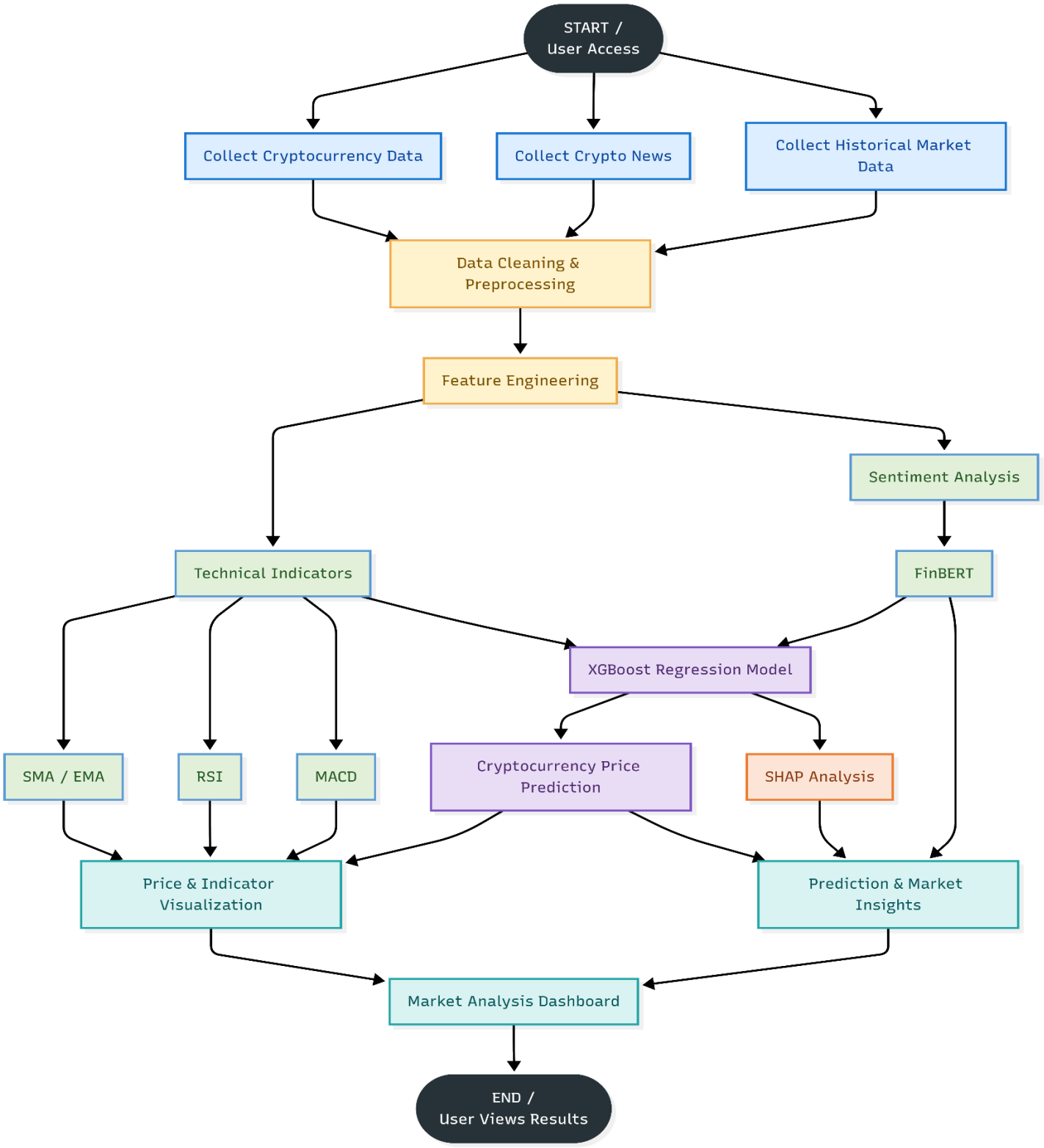


Figure 6.1 — Project Flow Analysis

7) Outcome of the Proposed Project

- To develop a system for predicting cryptocurrency prices
- To analyze sentiment related to cryptocurrency data
- To integrate prediction and sentiment analysis into a single platform
- To provide clear and simple visualization of results
- To reduce the need for manual analysis by users
- To improve decision-making using combined insights

8) Modules of the Proposed Project — Functional Requirements

The Cryptocurrency Prediction and Market Analysis System is organized into the following modules. Detailed, testable requirements for each are listed below with unique identifiers.

- **8.1 Data Ingestion & Storage**
- **8.2 Feature Engineering & Sentiment Analysis**
- **8.3 Machine Learning & Price Prediction**
- **8.4 Backend API & Watchlist Management**
- **8.5 Frontend Visualization & Dashboard**

Req. ID	Module	Description
FR-01	Data Ingestion	The system shall automatically fetch historical cryptocurrency prices (via CoinGecko) and news articles (via NewsAPI) and store them in the local database.
FR-02	Feature Engineering	The system shall calculate technical financial indicators including 7-day and 30-day moving averages (SMA/EMA), RSI, and volatility from the raw market data.
FR-03	Sentiment Analysis	The system shall process fetched cryptocurrency news articles using an NLP Transformer model to generate positive, negative, and neutral sentiment scores.

Req. ID	Module	Description
FR-04	Price Prediction	The system shall use a trained XGBoost machine learning model to predict the next-day closing price and expected percentage return.
FR-05	Watchlist	The system shall allow users to add or remove specific cryptocurrencies from a personalized watchlist.
FR-06	Dashboard UI	The system shall display historical price charts, predicted future prices, and recent news sentiment on a centralized web dashboard.
FR-07	Model Explainability	The system shall generate SHAP (SHapley Additive exPlanations) values to explain which features most influenced the day's prediction.
FR-08	REST API	The backend shall provide API endpoints for the frontend to request market data, predictions, and news securely.

Table 8.1 — Functional Requirements

9) Non-Functional Requirements of the Project

Req. ID	Module	Description
NFR-01	Performance	The frontend dashboard shall load and render interactive cryptocurrency price charts in under 3 seconds.
NFR-02	Scalability	The database and machine learning pipeline shall be modular enough to easily add support for new cryptocurrencies without requiring a schema change.

Req. ID	Module	Description
NFR-03	Reliability	If the external NewsAPI is temporarily unavailable, the prediction model shall safely fallback to using only technical indicators without failing completely.
NFR-04	Accuracy	The XGBoost prediction model shall aim to minimize the Mean Absolute Error (MAE) and correctly predict the directional movement of the asset.
NFR-05	Maintainability	The backend shall follow a decoupled architecture separating data fetching scripts, model inference, and REST APIs.
NFR-06	Usability	The web application shall be fully responsive, ensuring charts and predictions are clearly readable on both desktop and mobile devices.

Table 9.1 — Non-Functional Requirements

10) Project Deliverables

S.No.	Deliverable
1	Software Requirement Specification (SRS) Document
2	System Design Documentation (ER Diagram, Architecture Diagram, Database Schema)
3	Working Prototype — Cryptocurrency Prediction web application (React frontend, Flask backend, Python ML pipeline)
4	SQLite database (app.db) populated with historical market data, news, and generated ML features
5	User Manual / README with setup instructions to run the data pipelines and start the servers
6	Final Project Report and Presentation

Table 10.1 — Project Deliverables

11) Technology to be Used

The technologies below were learnt during the MCA programme (Python, database management, web development) and are applied to Cryptocurrency Prediction and Market Analysis System based on the rationale in the table.

Layer	Technology & Rationale
Frontend	React.js (with Vite) and Tailwind CSS — a modern, component-based UI chosen for lightning-fast development, utility-first styling, and building interactive cryptocurrency charts.
Backend	Python with Flask — a lightweight, micro-framework REST API architecture chosen for its seamless integration with the Python data science and machine learning pipelines.
Database	SQLite — a serverless, file-based relational database (app.db) that handles both structured market prices and ML feature sets, avoiding the operational overhead of a separate database server.
AI / ML	XGBoost and Hugging Face Transformers — XGBoost provides highly accurate gradient boosting for time-series price prediction, while PyTorch-based transformers run NLP sentiment analysis on crypto news.
Data Ingestion	CoinGecko and NewsAPI — robust external data APIs accessed via Python requests, chosen for automated daily ingestion of historical cryptocurrency market charts and relevant financial news.
Version Control	Git and GitHub — used for source control and collaborative development between the two team members.

Table 11.1 — Technology Stack and Rationale

12) Timeline of Activities Planned / Completed

The project is organized into six sprints spanning July to October, shown below in table format.

Sprint	Duration	Planned Activities	Status
Sprint 1	Jul – 1st & 2nd week	Requirement Analysis & Design - finalize ML algorithms, ER model, and technology stack.	**Completed**
Sprint 2	Jul 3rd week – Aug 2nd week	Data Pipelines & Database — SQLite schema, CoinGecko/NewsAPI ingestion scripts, and Flask backend setup.	**Completed**
Sprint 3	Aug 2nd week – Sept 1st week	AI/ML & Feature Engineering — Technical indicators (SMA/RSI), NLP Sentiment Analysis, and XGBoost model training.	**In Progress**
Sprint 4	Sept 2nd – 4th week	Frontend Development — React/Vite UI, Recharts integration for market charts, and dashboard layout.	**Planned**
Sprint 5	Oct 1st week	Integration & Watchlist — Connect React frontend to Flask REST API, implement Watchlist management.	**Planned**
Sprint 6	Oct 2nd week	Testing, Documentation & Final Review — System testing and preparing final reports.	**Planned**

Table 12.1 — Sprint Plan

Signature of the Student

Signature of the Guide